Andrew Jeyabose Sundar, Ph.D.

Senior Computer Vision & ML Scientist

andrewj.research@gmail.com • +1 919 904 6103 • Chapel Hill, NC • LinkedIn • Google Scholar

90+	2,250+	h-index 22	Top 2%
Publications	Citations	Researcher Impact	Stanford/Elsevier

Professional Summary

Computer Vision ML Scientist with 12+ years shipping deep learning models into production medical imaging pipelines. Architected contrastive learning and Vision Transformer systems that reached state-of-the-art on fMRI retrieval and tumor segmentation benchmarks. Led AI methodology for a $1.24M NIH R01 program and supervised 15+ researchers from prototype to peer-reviewed publication. Stanford/Elsevier Top 2% Scientist (2024, 2025). Known for converting research breakthroughs into containerized, GPU-optimized services — not just papers.

Core Technical Competencies

CV Architectures	Vision Transformers (ViT, Swin), CNNs (ResNet, EfficientNet), U-Net+Attention, SegFormer, hybrid models
Representation Learning	Contrastive learning (SimCLR, MoCo), self-supervised pre-training, metric learning, multimodal embedding fusion
Medical Imaging	Whole-slide histopathology (gigapixel WSI), fMRI/CT 3D-4D volumes, patch-based MIL, tile-based processing
Image Understanding	Semantic & instance segmentation, object detection (YOLO, Faster R-CNN), retrieval, biomarker discovery
Production ML	Distributed training (PyTorch DDP, Horovod), Docker, FastAPI, MLflow, W&B, TorchScript, quantization/pruning
Frameworks & Tools	PyTorch, TensorFlow, Hugging Face, PyTorch Geometric, PyTorch3D, scikit-learn, Git, CI/CD

Professional Experience

Research Associate / Senior Computer Vision Researcher | UNC Chapel Hill | Chapel Hill, NC*Dec 2024 – Present*

Technical AI lead for NIH R01NS125897 ($1.24M) — a gigapixel medical imaging program spanning fMRI, CT, and histopathology.

- Reduced distributed training time by 35% for terabyte-scale image datasets by re-engineering PyTorch DDP pipelines across multi-GPU clusters, enabling experiments that were previously cost-prohibitive.
- Delivered the project's first production-ready model suite — Vision Transformers + contrastive learning for 3D/4D fMRI — containerized with Docker/FastAPI and integrated with clinical retrieval workflows.
- Established MLOps standards (MLflow, W&B, versioned model registry) that cut researcher onboarding time and eliminated reproducibility gaps across a 6-person team.
- Mentored 2 Ph.D. students and 5 junior researchers; 3 mentees submitted first-author manuscripts within 12 months under direct guidance.
- Currently leading NIH K99/R00 proposal as Principal Investigator — translating lab CV methodology into an independent research program.

Research Faculty / Computer Vision Lead | Manipal Institute of Technology | Manipal, India*May 2022 – Dec 2024*

Built and led the institute's computer vision research group across 15 concurrent projects in medical and computational imaging.

- Published multiple peer-reviewed papers — spanning image segmentation, histopathology representation learning, and transformer-based classification.
- Achieved state-of-the-art classification and retrieval accuracy on BreakHis and BACH breast cancer histopathology benchmarks by designing SimCLR/MoCo contrastive pipelines with ViT backbones, outperforming prior supervised baselines on limited labeled data.
- Secured Co-PI role on ICMR grant proposal; authored the full technical design for a deep learning-based medical image analysis pipeline, including architecture, training protocol, and evaluation framework.
- Supervised 15+ undergraduate and master's students from literature review through publication; 8 student-led papers accepted at indexed journals or IEEE conferences.

Academic Mentor / Computer Vision Researcher | Karunya Institute of Technology and Sciences | Coimbatore, India
Jun 2013 – Apr 2022

9-year research and teaching track building CV expertise from CNNs to early transformer architectures.

- Produced mutiple publications on image classification, segmentation, and signal processing over nine years; won Best Paper Awards at three international conferences (2019, 2021, 2023).
- Built custom PyTorch implementations of ResNet, EfficientNet, and attention-augmented CNNs for medical imaging; established a reusable codebase that accelerated subsequent student projects by 2–3x.
- Elevated to IEEE Senior Member for sustained contributions to the computer vision and biomedical engineering community.

High-Impact Projects

Multimodal Contrastive Learning for fMRI Network Recovery | *Published: BioMedical Engineering OnLine 2025*

- Designed a SimCLR-based contrastive framework fusing rs-fMRI imaging with clinical measurements into a unified embedding space; achieved state-of-the-art on whole-brain network retrieval and classification across a multi-site clinical cohort.
- Engineered a custom data augmentation strategy for 4D fMRI volumes (temporal cropping, regional masking, signal perturbation) to generate contrastive pairs without label leakage — a key methodological contribution absent from prior neuroimaging contrastive work.
- Implemented distributed training across 8 GPUs with a custom embedding extraction and indexing pipeline; the resulting embedding store now serves as the core inference engine for all NIH R01 downstream classification and retrieval tasks.

Interpretable Vision Transformers for Epilepsy Classification | *Under review: Scientific Reports*

- Adapted Swin Transformer and ViT-Base architectures for volumetric 3D/4D fMRI processing by introducing factorized spatiotemporal attention and patch embedding strategies tailored to the irregular geometry of brain parcellation; overcame scarce clinical labels via masked autoencoder (MAE) self-supervised pre-training on unlabeled scans.
- Integrated gradient-weighted attention rollout to generate voxel-level saliency maps, enabling neuroscientists to identify epileptogenic network signatures without accessing model internals; maps validated against established neurological biomarkers by clinical co-investigators at UNC.
- Integrated attention map interpretability to surface neurological biomarkers — satisfying clinical collaborator requirements for model transparency in a diagnostic context; manuscript under review at Scientific Reports.

Self-Supervised Histopathology Embeddings for WSI Analysis | *BreakHis & BACH Datasets*

- Built a fully automated end-to-end pipeline for gigapixel whole-slide image analysis: tissue-aware tiling (256x256 patches at 20x magnification), stain normalization (Macenko method), SimCLR/MoCo contrastive pre-training on 2M+ unlabeled patches, and ABMIL aggregation for slide-level prediction — requiring zero manual labels at pre-training stage.

- Exceeded supervised ResNet-50 baselines on BreakHis 8-class breast cancer subtype classification and patch-level retrieval using only 10% of available labels at fine-tuning, demonstrating a practical path to clinical deployment in label-scarce histopathology settings.
- Delivered reusable PyTorch codebase adopted across 3 subsequent student-led histopathology projects; reduced new project setup time from weeks to days.

Transformer-Based Tumor Segmentation | *Published: Journal of Computational Biology 2024*

- Designed a hybrid SegFormer + attention-augmented U-Net architecture for glioma segmentation in multi-modal MRI (T1, T1ce, T2, FLAIR); introduced non-subsampled shearlet transform as a preprocessing layer to enhance tumor boundary detail, yielding measurable improvement in Hausdorff distance over standard preprocessing.
- Benchmarked against BraTS challenge baselines; model achieved competitive Dice scores on whole tumor, tumor core, and enhancing tumor sub-regions, with inference latency suitable for clinical review workflows.
- Published in Journal of Computational Biology (2024); code structured as a modular PyTorch library to support plug-and-play backbone swapping for future architecture comparisons.

Selected Publications — 90 Total | 2,250+ Citations | h-index 22

1. **Jeyabose A.** et al. "Multimodal contrastive learning on rs-fMRI to quantify whole-brain network recovery." BioMedical Engineering OnLine 24(1):125, 2025. *(Contrastive learning; multimodal embeddings; image retrieval)*
2. Jeyabose A. et al. "Interpretable Transformer Models for rs-fMRI Epilepsy Classification and Biomarker Discovery." medRxiv 2025; under review Scientific Reports. *(ViT; attention; biomarker discovery)*
3. Robinson B., **Jeyabose A.** et al. "A multimodal deep learning framework for functional brain network classification in rs-fMRI." Cognitive Neurodynamics 19(1):1-13, 2025. *(Multimodal DL; network classification)*
4. Tamilarasi M., **Jeyabose A.*** et al. "Detection and Segmentation of Glioma Tumors Utilizing a UNet CNN with Non-Subsampled Shearlet Transform." J. Computational Biology 31(8):757-768, 2024. *(Segmentation; medical imaging)*
5. Gunasekaran H., **Jeyabose A.*** et al. "GIT-Net: Ensemble Deep Learning for GI Tract Endoscopic Image Classification." Bioengineering 10(7):809, 2023. *(Image classification; ensemble DL)*
6. Sharafudeen M., **Jeyabose A.*** et al. "Leveraging Vision Attention Transformers for Dermoscopic Deepfake Detection." Diagnostics 13(5):825, 2023. *(ViT; deepfake detection; dermatology)*

** Senior / Corresponding Author*

Professional Service & Recognition

Editorial Roles	Associate Editor: Scientific Reports (Nature), PLOS ONE, Frontiers in AI; Guest Editor: Digital Pathology — Scientific Reports, Biomolecules (MDPI)
Peer Review	200+ manuscripts reviewed for IEEE, Elsevier, Springer, Nature, MDPI
Mentorship	Mentor, Neuromatch Academy Impact Scholar Program (2024–2025) — guided international early-career researchers in ML for science
Awards	Best Paper Awards: ICASE 2023, ICBDML 2021, ICACCS 2019; IEEE Senior Member

Education

Ph.D., Computer Science / Engineering (2016 – 2021) — Vellore Institute of Technology (VIT), India

Specialization: Privacy Preserving, Electronic Healthcare Records, Deep Learning, Computer Vision, Medical Image Analysis

M.E. / M.Tech., Computer Science and Engineering (2011 – 2013) — Anna University, India

Specialization: Data Mining, Machine Learning

B.E. / B.Tech., Computer Science and Engineering (2008 – 2011) — Anna University, India

Specialization: Networks and Communications